

10/18
11/20

Convert the following Expressions to Postfix.

1) $a + b * (c \wedge d - e) \wedge (f + g * h) - ?$

2) $(A * B) - (C - D)$

3) $(A * B) \wedge (C / D)$

Sol :-

$a \rightarrow$ Operand, so add to result.

1) Char -c $+$ $\rightarrow$ push to stack (top++)

0 a

$b \rightarrow$ add to result.

1 +

$* \rightarrow \text{top}! = -1 \& * > +$

2 b

$\rightarrow$ push.

3 *

$(\rightarrow$ push.

4 (

$c \rightarrow$ add to result.

5 c

$\wedge \rightarrow 3 < -1 F \Rightarrow$ push.

6 ^

$d \rightarrow$ add to result.

7 d

$- \rightarrow 1 < 3 T \Rightarrow$ pop $\wedge$ add it to result.

8 -

then, push -

9 e

$e \rightarrow$ add to result.

10)

$) \rightarrow$ pop -, add it to result.
remove (.

11 ^

$\wedge \rightarrow 3 < 2 F \Rightarrow$ push

12 (

$(\rightarrow$ push

13 f

$f \rightarrow$ add to result.

14 +

$+$ $\rightarrow 1 < -1 F \Rightarrow$ push.

15 g

$g \rightarrow$ add to result.

16 *

$* \rightarrow * > + \Rightarrow$ push.

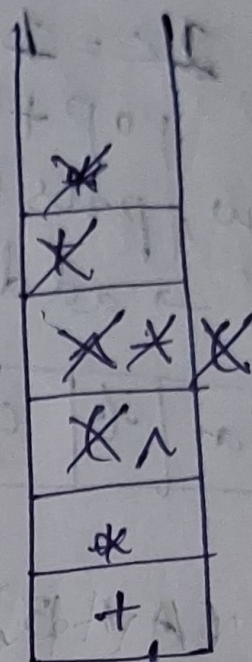
17 h

$h \rightarrow$ add to result.

18)

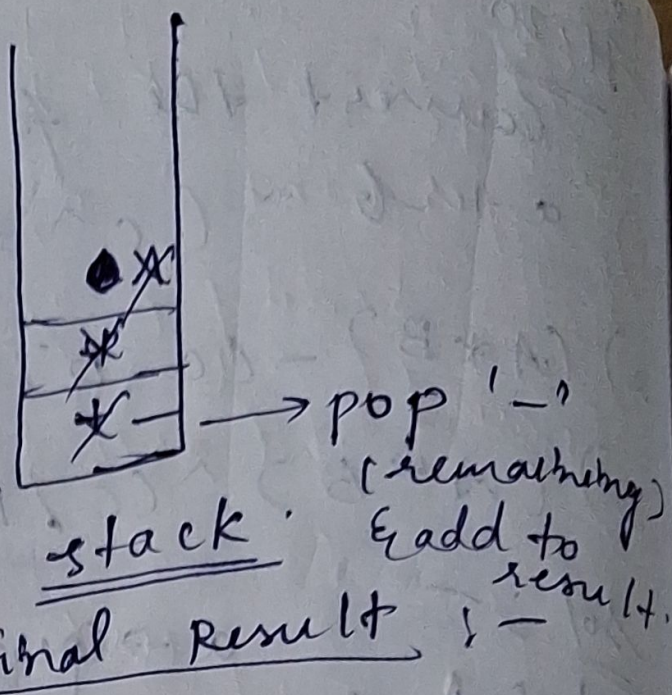
19 -

20



Stack

$\rightarrow$ pop $\times$ } Add to result
 pop + }
 remove (



$\rightarrow 1 < 3 T$

pop ^

$1 < 2 T$

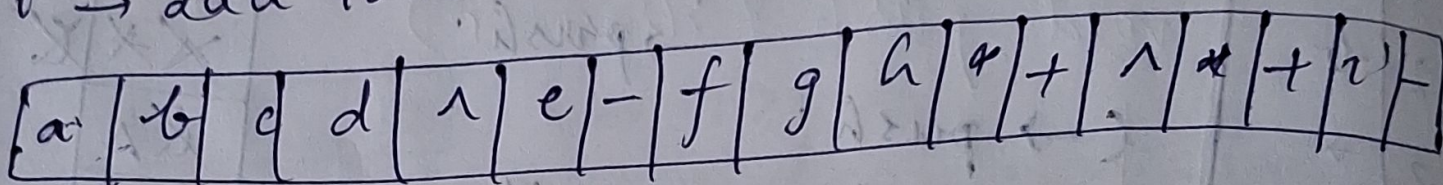
pop $\times$

$1 < 1 T$

pop +

Now, push -

$\rightarrow$ add to result.



2) $(A * B) - (C - D)$

i char-c

c - add to stack

0 (

A - add to result

1 A

2 $\times$

$\times$ - top! = -1 T

3 B

$2 < -1 F$

4)

$\Rightarrow$ add to stack

5 -

B - add to result

6 (

) - top! = -1 T

7 c

stack[top]! = 'c' T

8 -

add top to result.

9 D

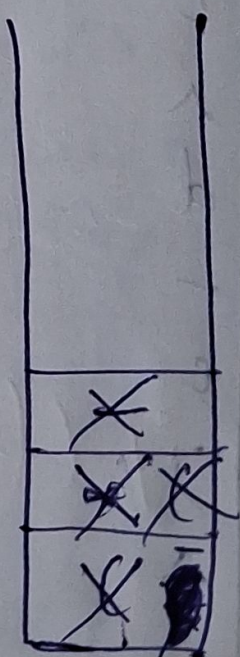
& remove 'c'.

10)

~~$\rightarrow$ add to result~~ push h

~~c $\rightarrow$ top! = -1 T~~ & push to stack

~~$1 < -1 T$~~
~~add top to result~~



c → add to result.
 - → 1 < = 1 F
 ⇒ push to stack.

Finally, pop the remaining elements in stack, i.e., '-'.
 Result :-

D → add to result.

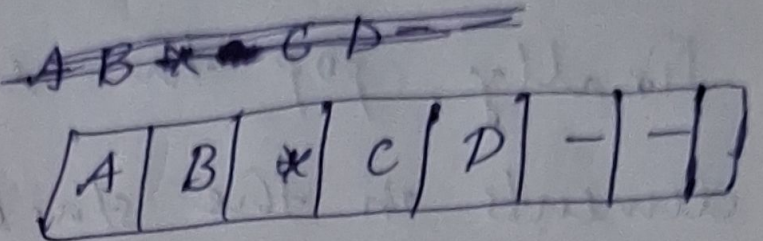
) → top! = -1

stack[top]! = 'c'

so, pop - & add to result.

stack[top] = 'c',

top - -



3) (A * B) ^ (C / D)

i char-c

0 (

1 A

2 *

3 B

4)

5 ^

6 (

7 C

8 /

9 D

10)

'(' → ~~push to stack~~

⇒ push to stack

A → add to result

* → top! = -1 T &
 2 < = -1 F

⇒ push to stack

B → add to result

) → top! = -1 &

stack[top]! = 'c' T

so, pop * & add to result

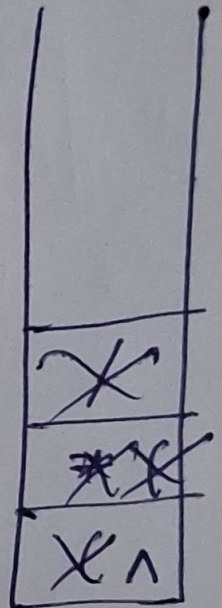
remove 'c'.

^ → top! = -1 F ⇒ push

(→ push

C → add to result

/ → top! = -1 T, 2 < = -1 F ⇒ push



stack

